SE 216 – SOFTWARE PROJECT MANAGEMENT
SOFTWARE TOOLS DOCUMENT

PROJECT NAME: Happy Tails

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TASK #	PROJECT TASKS WHICH REQUIRE SOFTWARE TOOL SUPPORT
1	Database System
2	AI Integration Service
3	Live Streaming
4	Payment Gateway

SOFTWARE TOOLS FOR TASK 1: Database System

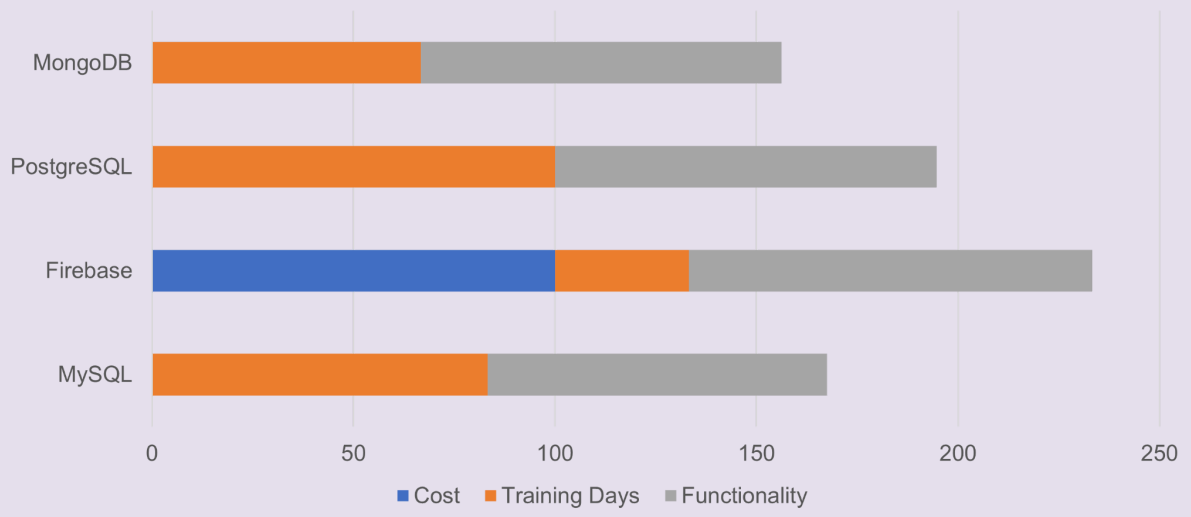
Tool Cost/Training/Functionality Data

Tool	MySQL	Firebase	PostgreSQL	MongoDB
Cost	$0	$25	$0	$0
Training Days	5 days	2 days	6 days	4 days
Functionality	80	95	90	85

Normalized Cost/Training/Functionality Data

Tool	MySQL	Firebase	PostgreSQL	MongoDB
Cost	0.0	100.0	0.0	0.0
Training Days	83.3	33.3	100.0	66.7
Functionality	84.2	100.0	94.7	89.5

Normalized Tool Graph



Which tool has been selected? Why?

PostgreSQL was selected as the database system for our project. Although it requires the longest training period of 6 days (100.0 normalized), it is completely free ($0 cost / 0.0 normalized), making it highly cost-effective compared to Firebase. Furthermore, its excellent functionality score (90 / 94.7 normalized) and robust relational database capabilities make it the most suitable choice for securely managing our complex, structured data, such as detailed animal profiles, health trackers, and item-specific donations.

SOFTWARE TOOLS FOR TASK 2: AI Integration Service (For Compatibility Matching)

Tool Cost/Training/Functionality Data

Tool	OpenAI API	Google Vertex AI	AWS Bedrock	Hugging Face API
Cost (per 1M tokens)	$20	$15	$18	$30
Training Days	5	10	12	7
Functionality	95	90	85	80

Normalized Cost/Training/Functionality Data

Tool	OpenAI API	Google Vertex AI	AWS Bedrock	Hugging Face API
Cost	66.7	50.0	60.0	100.0
Training Days	41.7	83.3	100.0	58.3
Functionality	100.0	94.7	89.5	84.2

Normalized Tool Graph

Which tool has been selected? Why?

OpenAI API was selected for our AI-Driven Compatibility Matching feature. Although it has a moderate cost of $20 for 1M tokens, it has the highest functionality score of 100.0 normalized for developing our complex scoring algorithm. Moreover, it has comprehensive documentation, which gives our project a very manageable learning curve with an average of 5 days or 41.7 normalized. This will enable our team to integrate the survey-matching system without compromising our project delivery time.

SOFTWARE TOOLS FOR TASK 3: Live Streaming

Tool Cost/Training/Functionality Data

Tool	WebRTC	Agora	AWS IVS	YouTube Live API
Cost (per 1k min)	$0	$0.59	$2	$0
Training Days	15	5	8	3
Functionality	85	95	90	75

Normalized Cost/Training/Functionality Data

Tool	WebRTC	Agora	AWS IVS	YouTube Live API
Cost	0.0	29.5	100.0	0.0
Training Days	100.0	33.3	53.3	20.0
Functionality	89.5	100.0	94.7	78.9

Normalized Tool Graph

Which tool has been selected? Why?

Agora was selected for providing the best balance between cost and performance. While it has a cost of $0.59 per 1k minutes, it delivers a high functionality score for low-latency streaming. Additionally, its short training period of 5 days allows our team to implement the live feed feature rapidly, ensuring we stay on schedule for other critical development tasks.

SOFTWARE TOOLS FOR TASK 4: Payment Gateway

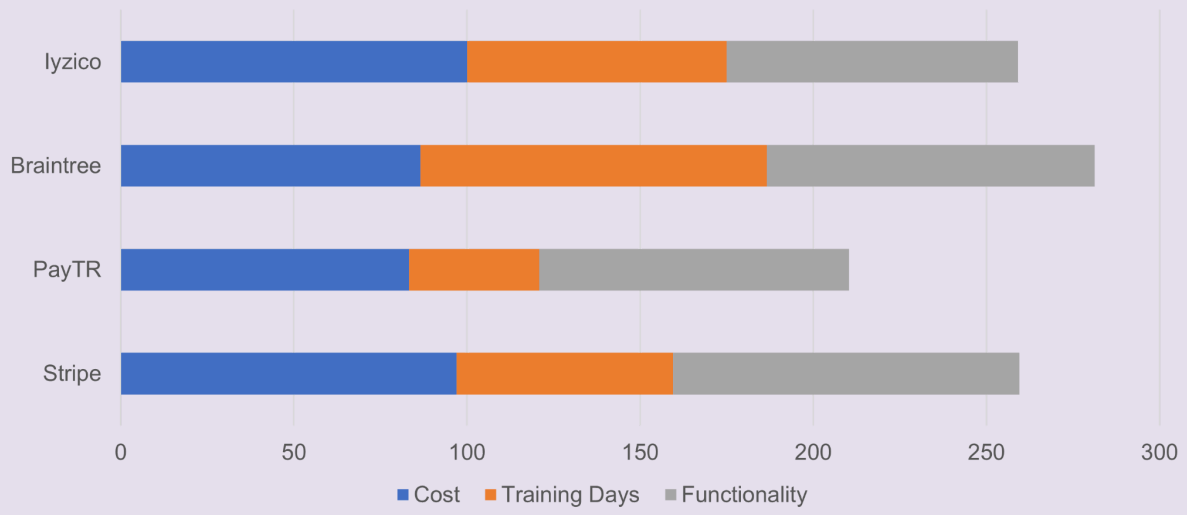
Tool Cost/Training/Functionality Data

Tool	Stripe	PayTR	Braintree	Iyzico
Cost (Transaction Fee %)	2.90	2.49	2.59	2.99
Training Days	5	3	8	6
Functionality	95	85	90	80

Normalized Cost/Training/Functionality Data

Tool	Stripe	PayTR	Braintree	Iyzico
Cost	97.0	83.3	86.6	100.0
Training Days	62.5	37.5	100.0	75.0
Functionality	100.0	89.5	94.7	84.2

Normalized Tool Graph



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Which tool has been selected? Why?

Stripe is chosen because it completely meets our stringent PCI DSS compliance criteria for the direct adoption and donation tool. It requires a slightly longer training period than the local solution, PayTR (5 days / 62.5 normalized), but its higher functionality score (100.0 normalized) and widely accepted and developer-friendly API make it the strongest option available. It guarantees that we can process financial transactions seamlessly and securely without ever storing the credit card details of the users in our database.